

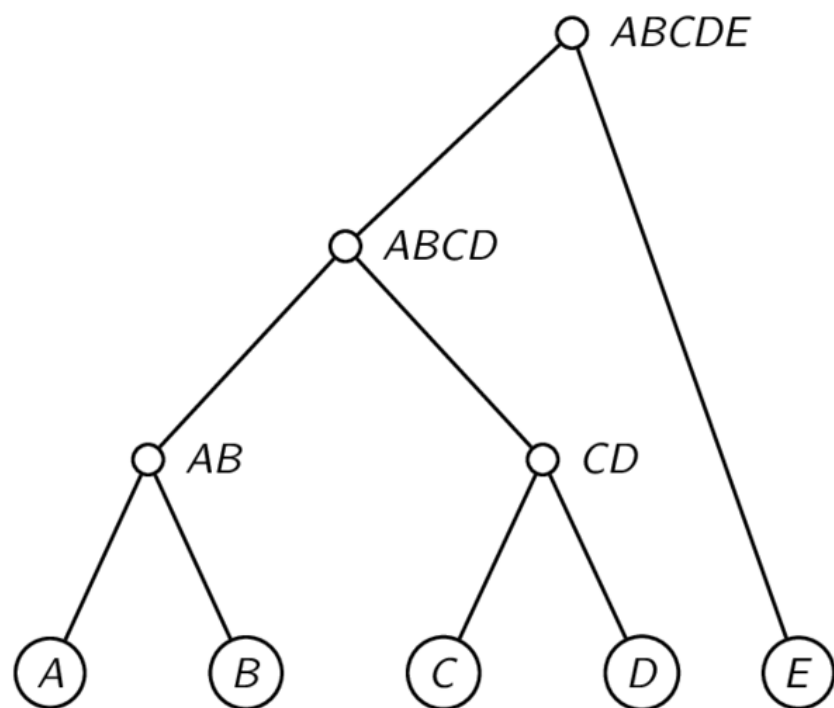
Hierarchical Clustering

The goal of hierarchical clustering is to create a sequence of nested partitions, which can be conveniently visualized via a tree or hierarchy of clusters, also called the cluster *dendrogram*.

The clusters in the hierarchy range from the fine-grained to the coarse-grained – the lowest level of the tree (the leaves) consists of each point in its own cluster, whereas the highest level (the root) consists of all points in one cluster.

Agglomerative hierarchical clustering methods work in a bottom-up manner. Starting with each of the n points in a separate cluster, they repeatedly merge the most similar pair of clusters until all points are members of the same cluster.

Hierarchical Clustering Dendrogram



The dendrogram represents the following sequence of nested partitions:

Clustering	Clusters
$\mathcal{C}_1$	$\{A\}, \{B\}, \{C\}, \{D\}, \{E\}$
$\mathcal{C}_2$	$\{AB\}, \{C\}, \{D\}, \{E\}$
$\mathcal{C}_3$	$\{AB\}, \{CD\}, \{E\}$
$\mathcal{C}_4$	$\{ABCD\}, \{E\}$
$\mathcal{C}_5$	$\{ABCDE\}$

with $\mathcal{C}_{t-1} \subset \mathcal{C}_t$ for $t = 2, \dots, 5$. We assume that A and B are merged before C and D .

Hierarchical Clustering: Nested Partitions

Given a dataset $\mathbf{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, where $\mathbf{x}_i \in \mathbb{R}^d$, a clustering $\mathcal{C} = \{C_1, \dots, C_k\}$ is a partition of $\mathbf{D}$.

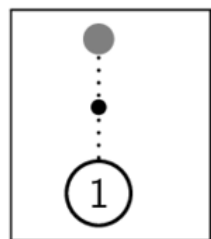
A clustering $\mathcal{A} = \{A_1, \dots, A_r\}$ is said to be nested in another clustering $\mathcal{B} = \{B_1, \dots, B_s\}$ if and only if $r > s$, and for each cluster $A_i \in \mathcal{A}$, there exists a cluster $B_j \in \mathcal{B}$, such that $A_i \subseteq B_j$.

Hierarchical clustering yields a sequence of n nested partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$. The clustering $\mathcal{C}_{t-1}$ is nested in the clustering $\mathcal{C}_t$. The cluster dendrogram is a rooted binary tree that captures this nesting structure, with edges between cluster $C_i \in \mathcal{C}_{t-1}$ and cluster $C_j \in \mathcal{C}_t$ if C_i is nested in C_j , that is, if $C_i \subset C_j$.

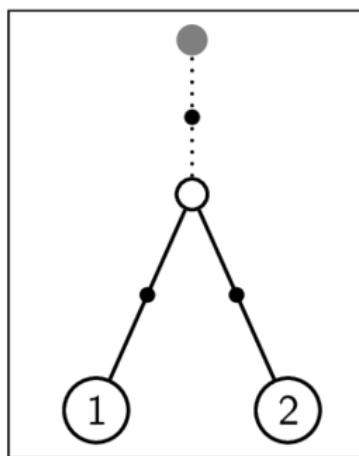
Number of Hierarchical Clusterings

The total number of different dendrograms with n leaves is given as:

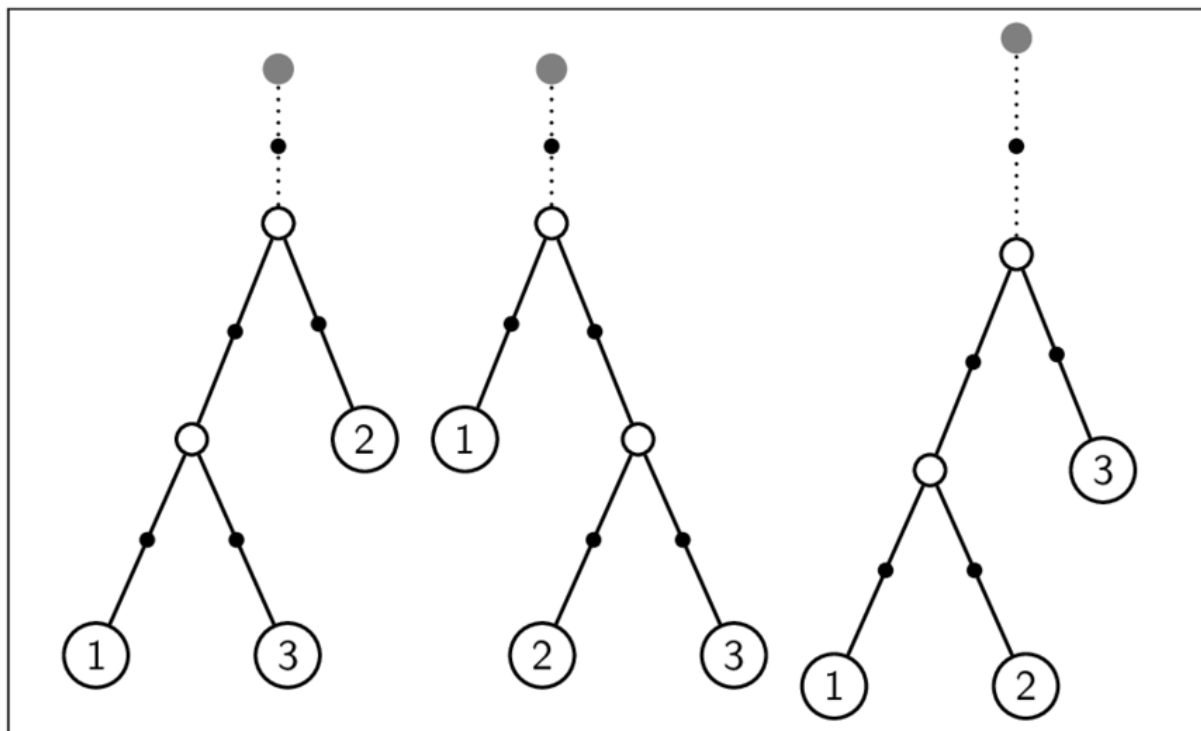
$$\prod_{m=1}^{n-1} (2m-1) = 1 \times 3 \times 5 \times 7 \times \cdots \times (2n-3) = (2n-3)!!$$



(a) $n=1$



(b) $n=2$



(c) $n=3$

Agglomerative Hierarchical Clustering

In agglomerative hierarchical clustering, we begin with each of the n points in a separate cluster. We repeatedly merge the two closest clusters until all points are members of the same cluster.

Given a set of clusters $\mathcal{C} = \{C_1, C_2, \dots, C_m\}$, we find the *closest* pair of clusters C_i and C_j and merge them into a new cluster $C_{ij} = C_i \cup C_j$.

Next, we update the set of clusters by removing C_i and C_j and adding C_{ij} , as follows $\mathcal{C} = (\mathcal{C} \setminus \{C_i, C_j\}) \cup \{C_{ij}\}$.

We repeat the process until $\mathcal{C}$ contains only one cluster. If specified, we can stop the merging process when there are exactly k clusters remaining.

Agglomerative Hierarchical Clustering Algorithm

AgglomerativeClustering(D, k):

- 1 $\mathcal{C} \leftarrow \{C_i = \{\mathbf{x}_i\} \mid \mathbf{x}_i \in \mathbf{D}\}$ // Each point in separate cluster
- 2 $\Delta \leftarrow \{\delta(\mathbf{x}_i, \mathbf{x}_j) : \mathbf{x}_i, \mathbf{x}_j \in \mathbf{D}\}$ // Compute distance matrix
- 3 **repeat**
- 4 Find the closest pair of clusters $C_i, C_j \in \mathcal{C}$
- 5 $C_{ij} \leftarrow C_i \cup C_j$ // Merge the clusters
- 6 $\mathcal{C} \leftarrow (\mathcal{C} \setminus \{C_i, C_j\}) \cup \{C_{ij}\}$ // Update the clustering
- 7 Update distance matrix Δ to reflect new clustering
- 8 **until** $|\mathcal{C}| = k$

Distance between Clusters

Single, Complete and Average

A typical distance between two points is the Euclidean distance or L_2 -norm

$$\|\mathbf{x} - \mathbf{y}\|_2 = \left(\sum_{i=1}^d (x_i - y_i)^2 \right)^{1/2}$$

Single Link: The minimum distance between a point in C_i and a point in C_j

$$\delta(C_i, C_j) = \min\{\|\mathbf{x} - \mathbf{y}\| \mid \mathbf{x} \in C_i, \mathbf{y} \in C_j\}$$

Complete Link: The maximum distance between points in the two clusters:

$$\delta(C_i, C_j) = \max\{\|\mathbf{x} - \mathbf{y}\| \mid \mathbf{x} \in C_i, \mathbf{y} \in C_j\}$$

Group Average: The average pairwise distance between points in C_i and C_j :

$$\delta(C_i, C_j) = \frac{\sum_{\mathbf{x} \in C_i} \sum_{\mathbf{y} \in C_j} \|\mathbf{x} - \mathbf{y}\|}{n_i \cdot n_j}$$

Distance between Clusters: Mean and Ward's

Mean Distance: The distance between two clusters is defined as the distance between the means or centroids of the two clusters:

$$\delta(C_i, C_j) = \|\mu_i - \mu_j\|$$

Minimum Variance or Ward's Method: The distance between two clusters is defined as the increase in the sum of squared errors (SSE) when the two clusters are merged, where the SSE for a given cluster C_i is given as

$$\delta(C_i, C_j) = \Delta SSE_{ij} = SSE_{ij} - SSE_i - SSE_j$$

where $SSE_i = \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \mu_i\|^2$. After simplification, we get:

$$\delta(C_i, C_j) = \left(\frac{n_i n_j}{n_i + n_j} \right) \|\mu_i - \mu_j\|^2$$

Ward's measure is therefore a weighted version of the mean distance measure.

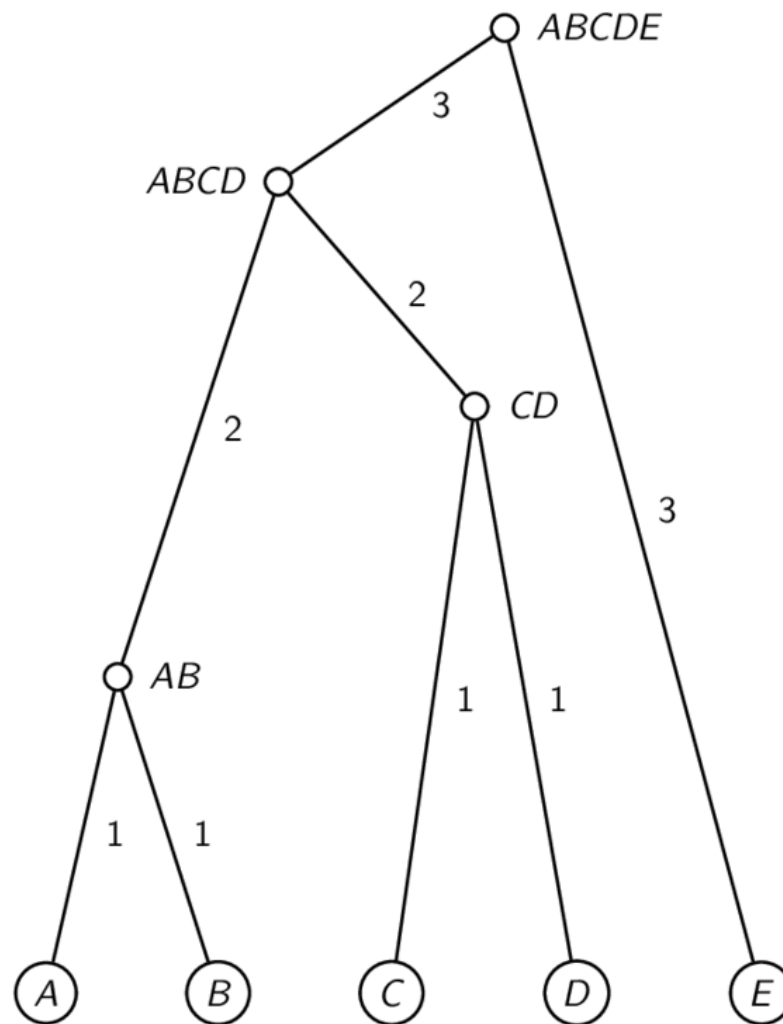
Single Link Agglomerative Clustering

δ	E
$ABCD$	(3)

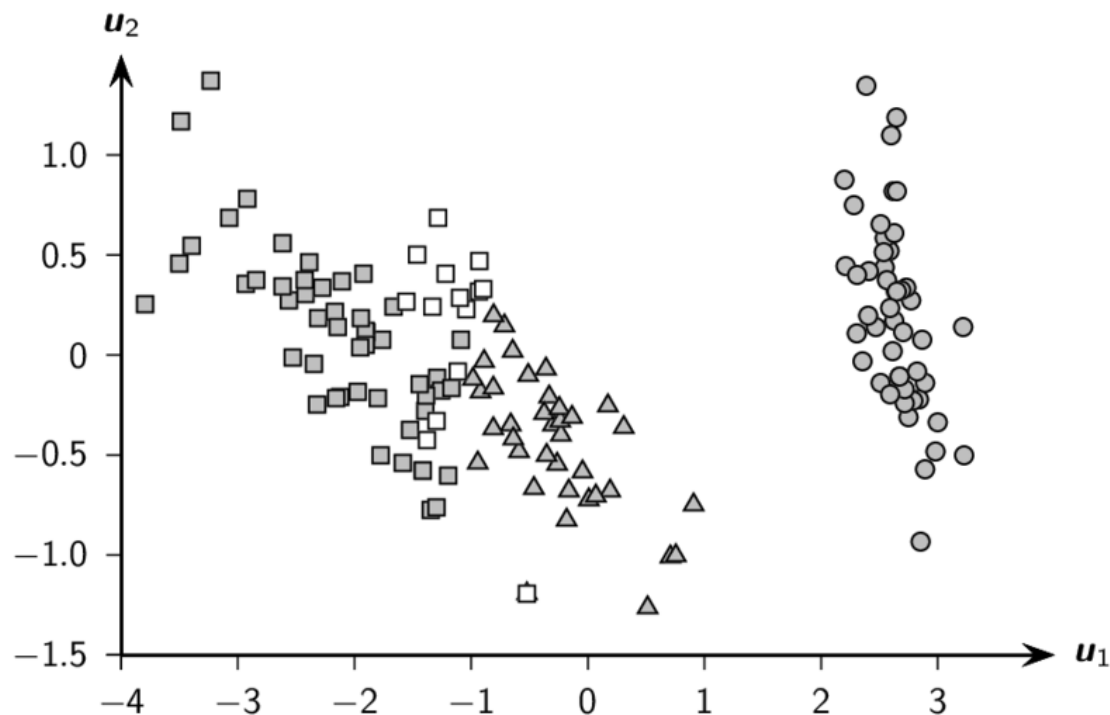
δ	CD	E
AB	(2)	3
CD		3

δ	C	D	E
AB	3	2	3
C		(1)	3
D			5

δ	B	C	D	E
A	(1)	3	2	4
B		3	2	3
C			1	3
D				5



Iris Dataset: Complete Link Clustering



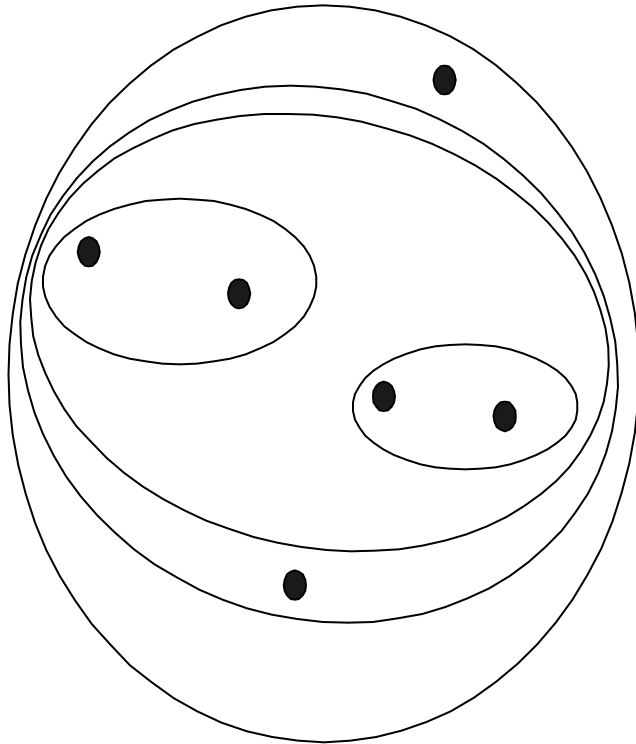
Contingency Table:

	iris-setosa	iris-virginica	iris-versicolor
C_1 (circle)	50	0	0
C_2 (triangle)	0	1	36
C_3 (square)	0	49	14

Hierarchical Clustering: Questions

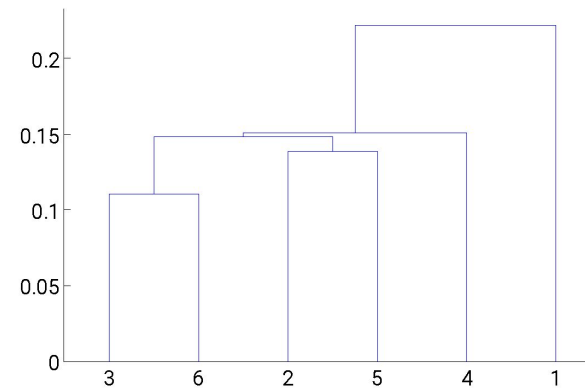
- What is the goal of Hierarchical Clustering?
- How do you represent a hierarchical clustering?
- What is nested partition, explain with an example.
- What are the different types of HCs?
- Write down the algorithm for Agglomerative clustering, in which fashion does it work?
- What are the different distance measures for inter clustering, show with examples?
- Explain single link agglomerative clustering with an example?

Hierarchical Clustering: MIN



Nested Clusters

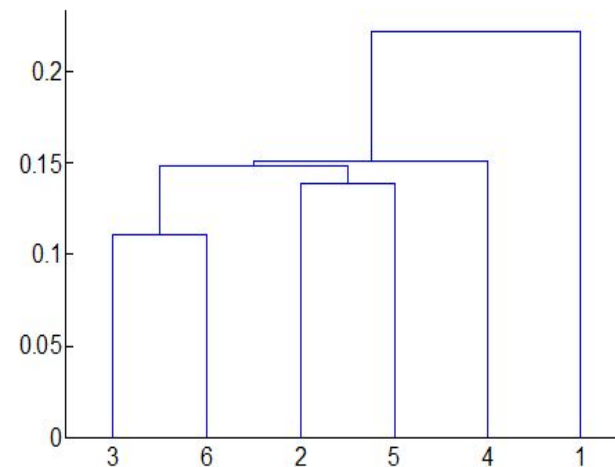
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00



Dendrogram

Distance between Clusters

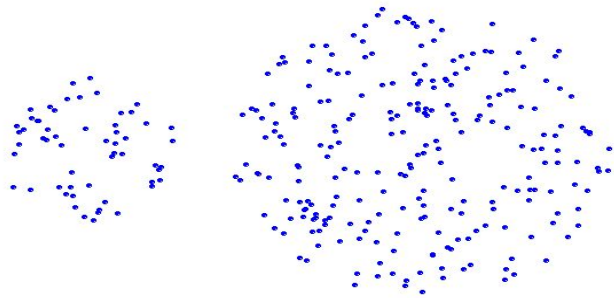
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00



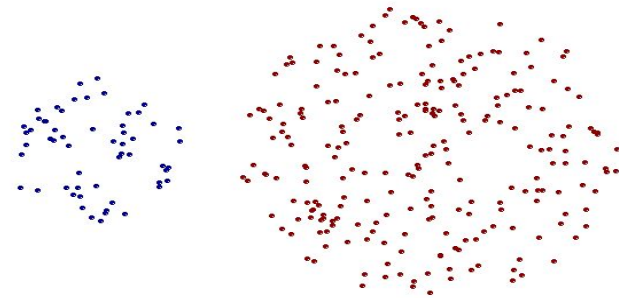
Dendrogram

$$\begin{aligned}
 \text{dist}(\{3, 6\}, \{2, 5\}) &= \min(\text{dist}(3, 2), \text{dist}(6, 2), \text{dist}(3, 5), \text{dist}(6, 5)) \\
 &= \min(0.15, 0.25, 0.28, 0.39) \\
 &= 0.15.
 \end{aligned}$$

Strength of MIN



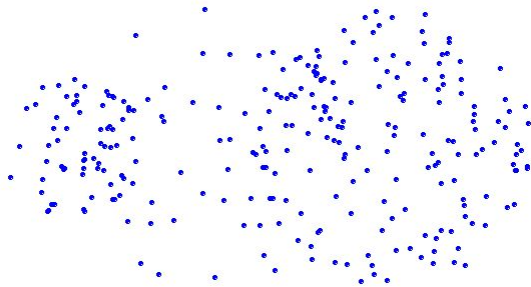
Original Points



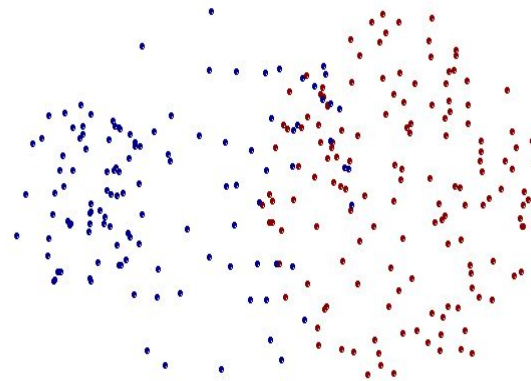
Two Clusters

- Can handle non-elliptical shapes

Limitations of MIN



Original Points

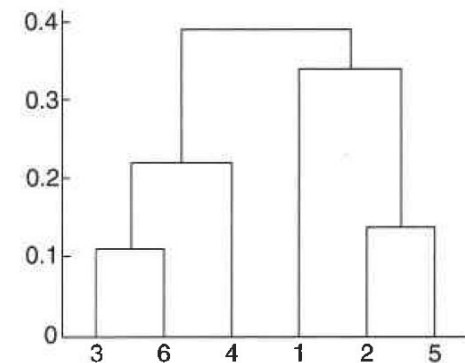
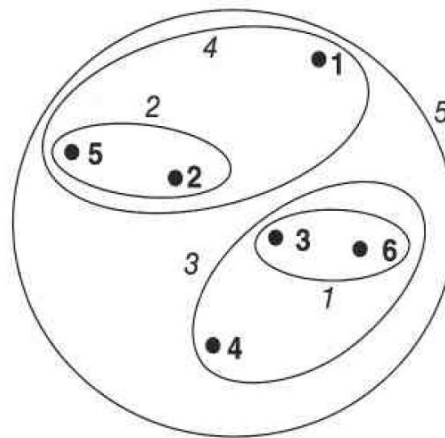


Two Clusters

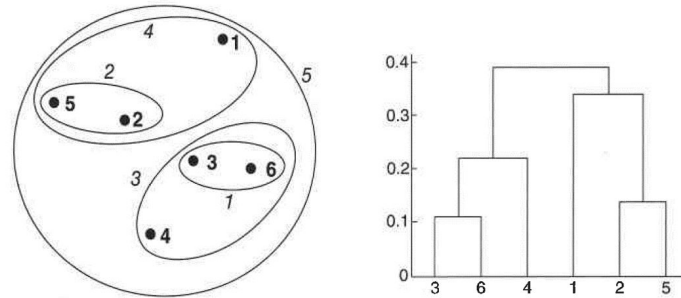
- Sensitive to noise and outliers

Cluster Similarity: MAX or Complete Linkage

- Similarity of two clusters is based on the two least similar (most distant) points in the different clusters
 - Determined by all pairs of points in the two clusters



Hierarchical Clustering: MAX



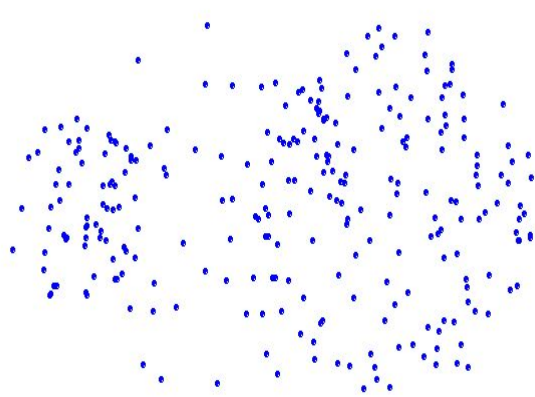
$$\begin{aligned}
 dist(\{3, 6\}, \{4\}) &= \max(dist(3, 4), dist(6, 4)) \\
 &= \max(0.15, 0.22) \\
 &= 0.22.
 \end{aligned}$$

$$\begin{aligned}
 dist(\{3, 6\}, \{2, 5\}) &= \max(dist(3, 2), dist(6, 2), dist(3, 5), dist(6, 5)) \\
 &= \max(0.15, 0.25, 0.28, 0.39) \\
 &= 0.39.
 \end{aligned}$$

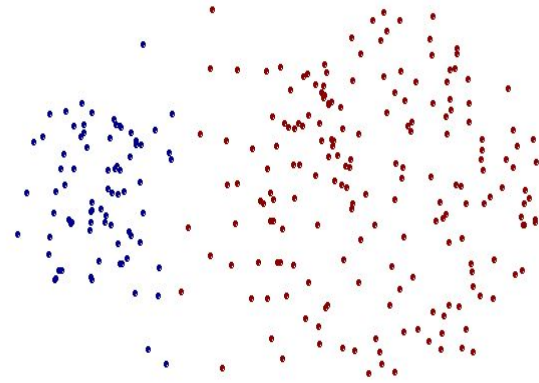
$$\begin{aligned}
 dist(\{3, 6\}, \{1\}) &= \max(dist(3, 1), dist(6, 1)) \\
 &= \max(0.22, 0.23) \\
 &= 0.23.
 \end{aligned}$$

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Strength of MAX



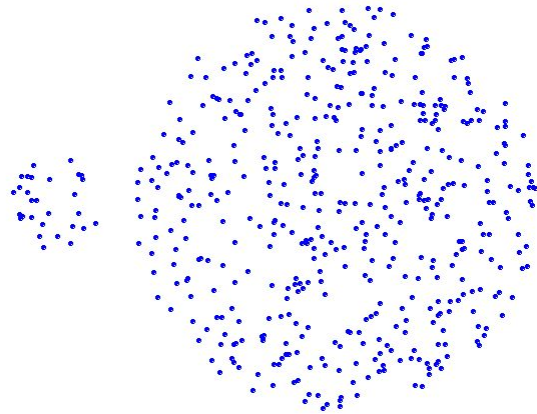
Original Points



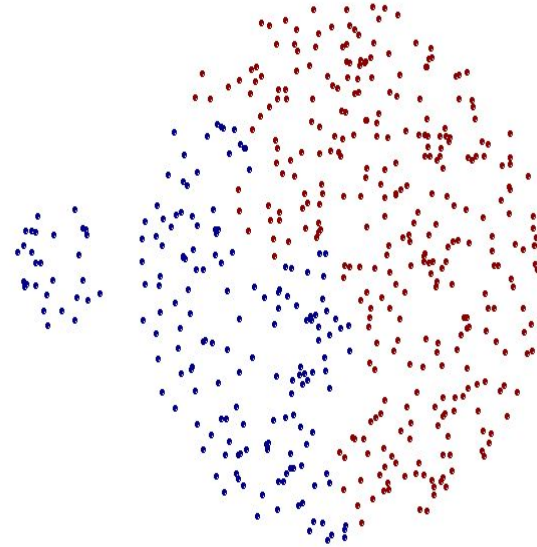
Two Clusters

- Less susceptible to noise and outliers

Limitations of MAX



Original Points



Two Clusters

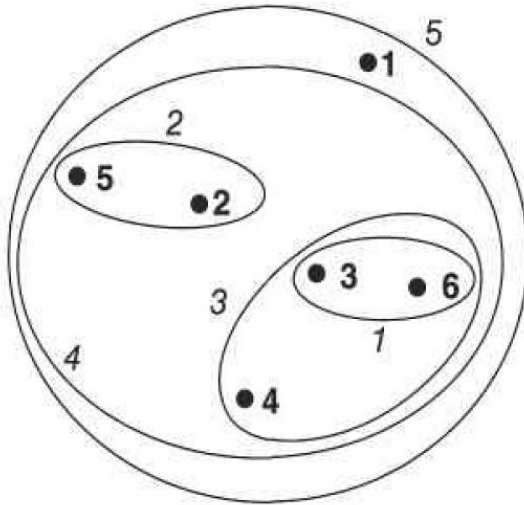
- Tends to break large clusters
- Biased towards globular clusters

Cluster Similarity: Group Average

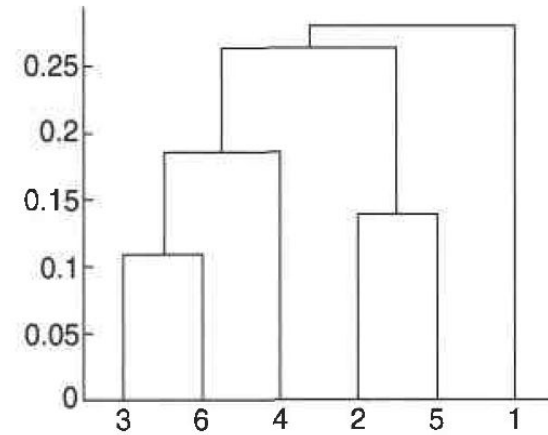
- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.
- Need to use average connectivity for scalability since total proximity favors large clusters

$$proximity(C_i, C_j) = \frac{\sum_{\substack{\mathbf{x} \in C_i \\ \mathbf{y} \in C_j}} proximity(\mathbf{x}, \mathbf{y})}{m_i * m_j}.$$

Hierarchical Clustering: Group Average



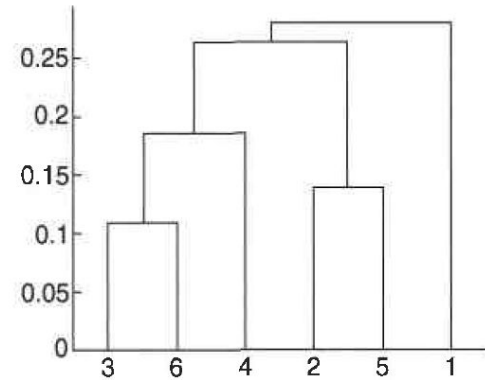
Nested Clusters



Dendrogram

Inter cluster distance: Group Average

$$proximity(C_i, C_j) = \frac{\sum_{\substack{\mathbf{x} \in C_i \\ \mathbf{y} \in C_j}} proximity(\mathbf{x}, \mathbf{y})}{m_i * m_j}.$$



$$\begin{aligned} dist(\{3, 6, 4\}, \{1\}) &= (0.22 + 0.37 + 0.23)/(3 * 1) \\ &= 0.28 \end{aligned}$$

$$\begin{aligned} dist(\{2, 5\}, \{1\}) &= (0.2357 + 0.3421)/(2 * 1) \\ &= 0.2889 \end{aligned}$$

$$\begin{aligned} dist(\{3, 6, 4\}, \{2, 5\}) &= (0.15 + 0.28 + 0.25 + 0.39 + 0.20 + 0.29)/(6 * 2) \\ &= 0.26 \end{aligned}$$

Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
 - Less susceptible to noise and outliers
- Limitations
 - Biased towards globular clusters

Jai Sairam

Data Mining and Machine Learning: Fundamental Concepts and Algorithms

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Mohammed J. Zaki¹ Wagner Meira Jr.²

¹Department of Computer Science
Rensselaer Polytechnic Institute, Troy, NY, USA

²Department of Computer Science
Universidade Federal de Minas Gerais, Belo Horizonte, Brazil

Chapter 14: Hierarchical Clustering

